Extracting Management Sentiment from Earnings Call Transcripts: A Calibrated LLM Pipeline and Its Limits

A technical report on the design, verification and honest evaluation of an end-to-end NLP system

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Abstract

Earnings call transcripts contain information that reported figures do not: hedged guidance, newly introduced caveats, and shifts in how management characterises demand. This report documents an end-to-end system that ingests a portfolio definition, retrieves recent earnings call transcripts for each holding, applies a large language model to extract a bounded sentiment score and a set of discussed themes, normalises the results onto a true calendar timeline, and presents the resulting trends through an interactive dashboard. The corpus comprises 24 transcripts (8 holdings x 3 quarters, approximately 190,000 words). Three findings are reported. First, prompt calibration was decisive rather than cosmetic: an initial scoring rubric assigned an identical score to three of four test holdings, and restructuring the rubric doubled the dispersion across holdings. Second, thematic extraction yielded a genuine cross-portfolio signal, with a single theme appearing in half of the holdings. Third, no statistically detectable relationship was found between sentiment change and subsequent price movement ($r = +0.30$, $n = 13$, $t = 1.04$). This null result is reported as the principal quantitative finding, together with the diagnosis that limits its strength: the scoring scale resolved to only five distinct values, leaving 9 of 13 observations with zero sentiment change. The study therefore measured the resolution of its own instrument more than it tested its hypothesis.

1. Introduction

Quarterly earnings calls are among the few occasions on which company executives speak at length, under questioning, about conditions in their business. The quantitative content of those calls is disseminated within seconds. The linguistic content is not: a transcript runs between roughly 5,000 and 11,000 words, and a diversified retail portfolio generates dozens of them per year. The practical consequence is that qualitative information which is public, material and freely available goes unread.

This project asks whether a large language model can compress that material into a comparable signal across holdings, and whether the resulting signal carries any predictive content. The deliberate emphasis throughout is on *verification*: each stage of the pipeline is accompanied by an automated check that must produce an explicit pass or fail result, and the evaluation reports what was found rather than what would have been convenient.

2. System Architecture

The system is implemented in Python 3.12 as six sequential stages, each isolated in its own module with a corresponding verification script.

Stage	Module	Responsibility
0 Setup	config.py	Credential loading; provider connectivity
1 Ingestion	csv_loader.py	Portfolio parsing with row-level validation
2 Retrieval	transcript_fetcher.py	Transcript acquisition and local caching
3 Extraction	llm_extract.py	LLM inference; schema validation
4 Aggregation	trend_aggregator.py	Temporal normalisation; theme counting
5 Presentation	app.py	Interactive dashboard
6 Evaluation	price_correlation.py	Sentiment-price correlation analysis

Table 1. Pipeline stages and their corresponding implementation modules.

Inference is performed by Llama 3.1 70B Instruct, accessed through NVIDIA's NIM endpoint using the OpenAI-compatible client. Transcripts are retrieved from AlphaVantage and cached on disk; because the free tier permits approximately 25 requests per day and a full portfolio run consumes 24, the cache is a functional requirement rather than an optimisation.

3. Methods

3.1 Extraction schema

Each transcript is submitted to the model with a system prompt specifying strict JSON output conforming to a fixed schema: a sentiment score bounded on [-1, 1], a categorical label, a list of two to five themes, a one-sentence statement of what shifted in management's emphasis, and a self-reported confidence level. Responses are validated before acceptance; an object failing any constraint is rejected and retried, and no partially populated result is ever returned to the caller. Inference temperature is set to 0.1, as the task is extraction rather than generation.

3.2 Calibration

The initial rubric partitioned the positive range into a single band spanning 0.4 to 1.0. Under this specification, three of four pilot holdings received a score of exactly 0.80. A dashboard constructed on such output would render a flat line across the portfolio and convey nothing, since the entire purpose of the instrument is to differentiate holdings from one another.

The rubric was accordingly restructured along two axes. The positive range was subdivided, with the uppermost band reserved for calls exhibiting unambiguous strength and no material caveat raised by management themselves. Separately, the prompt was amended to require that self-reported headwinds reduce the score even where headline results exceeded expectations. The effect is given in Table 2.

Holding	Initial rubric	Revised rubric
AAPL	0.80	0.65
NVDA	0.80	0.85

TSLA	0.80	0.55
PG	0.60	0.45
Dispersion (max - min)	0.20	0.40

Table 2. Effect of rubric revision on score dispersion. Dispersion across holdings doubled; the resulting ordering is economically defensible.

Two defects were exposed by this process. First, with a free-text label field the model began returning the rubric's own band descriptions verbatim as its characterisation of a call, text which would have propagated directly to the user interface; the label vocabulary was consequently closed to six values and enforced in validation. Second, score and label could disagree, with 0.80 arriving as *confident* while 0.70 arrived as *very confident*. The label is now derived deterministically from the score, with the model's own label retained separately for quality assurance.

3.3 Temporal normalisation

The transcript provider labels each call by the reporting company's own fiscal calendar. This is a material hazard for cross-sectional comparison, as Table 3 shows: a single label corresponds to periods spanning eleven months.

Holding	Provider label	Actual period end
NVIDIA	2026Q2	July 2025
Microsoft	2026Q2	December 2025
Apple	2026Q2	March 2026
JPMorgan	2026Q2	June 2026

Table 3. Identical provider labels corresponding to calendar periods up to eleven months apart.

All downstream ordering and plotting is therefore performed on the calendar month in which each fiscal period actually closed. Fiscal year-end months are maintained in an explicit lookup, as the provider returns no period-end date from which they could be derived; holdings absent from the lookup default to the calendar year and are reported rather than assumed silently.

4. Verification Methodology

Every stage carries an automated check printing an explicit pass or fail verdict, and a stage is not considered complete merely because its code executed without raising. Two checks are worth recording because both initially returned a passing verdict while the system was defective, and both failures share a structure.

(i) Authentication. Connectivity was initially verified by querying the provider's model catalogue. That endpoint responds without valid credentials, so an invalid key produced a passing result. The check was replaced with a minimal authenticated inference call that distinguishes rejection (401/403) from rate limiting (429) and from an unavailable model.

(ii) Dashboard rendering. The dashboard check confirmed that the server started, reported healthy and served a page of plausible size. All three held while the application was in fact raising an ImportError and rendering a traceback in place of the theme heatmap, because the framework writes exceptions into the page rather than to standard error. The check was replaced with one that executes the application through the framework's own test harness and inspects the rendered element tree.

The common lesson is that an absence of rejection is not evidence of correct function. Both defects were found by direct inspection of the running artefact, which argues for retaining manual observation alongside automated checks rather than treating the latter as a substitute.

5. Results

5.1 Sentiment trajectories

Table 4 presents the extracted sentiment for all eight holdings in true chronological order.

Holding	t-2	t-1	t	Change	Direction
AAPL	0.85	0.85	0.80	-0.05	flat
NVDA	0.85	0.85	0.80	-0.05	flat
MSFT	0.80	0.80	0.75	-0.05	declining
JPM	0.55	0.55	0.65	+0.10	improving
XOM	0.80	0.65	0.65	-0.15	declining
PG	0.55	0.55	0.55	0.00	flat
TSLA	0.65	0.65	0.65	0.00	flat
CRM	0.80	0.80	0.80	0.00	flat

Table 4. Sentiment by holding across three consecutive quarters, ordered by actual period end.

No holding exhibits two consecutive quarters of decline, the criterion adopted for the watch-list indicator. ExxonMobil records the single largest movement, declining once and then holding. This is reported as observed; no threshold was adjusted in order to generate an alert.

5.2 Thematic extraction

Thematic extraction proved the more informative component. The theme *AI adoption* appears in four of eight holdings spanning unrelated sectors, a portfolio-level observation that no individual call would surface. At the

word level, five holdings reference adoption, four reference investment, and three each reference constraints, expansion and supply. A word-level view is reported alongside exact theme matching because companies rarely phrase a shared trend identically; *AI capex*, *AI investments* and *AI adoption* denote one trend in three formulations, and exact matching alone understates genuine overlap.

Factual grounding was spot-checked against source transcripts. Both sampled claims in the *notable shift* field, concerning Apple's announced chief executive transition and its withdrawal of a net-cash-neutral capital structure target, were verified as near-verbatim and not fabricated.

5.3 Sentiment and subsequent price movement

The stretch hypothesis was that a change in extracted sentiment between consecutive calls would precede price movement in the same direction. Sentiment changes were paired with the holding's return over the subsequent ninety days, with the call date approximated as the fiscal period end plus thirty-five days.

Statistic	Value
Pearson correlation coefficient (r)	+0.2993
Observations (n)	13
t statistic	1.040
Two-sided 5% critical t (df = 11)	2.201
Directional agreement	6 of 13 (46%)
Observations with non-zero sentiment change	4 of 13 (31%)

Table 5. Correlation between sentiment change and subsequent 90-day return.

The coefficient does not approach the threshold for statistical significance, and directional agreement is marginally below the 50 percent expected from random assignment. **No detectable relationship was found.**

The diagnosis beneath that result is more informative than the coefficient. Nine of thirteen observations recorded a sentiment change of exactly zero, because the scoring resolved to only five distinct values across the entire corpus. An independent variable that is near-constant cannot correlate with anything. The experiment therefore characterised the resolution of the measurement instrument rather than testing the hypothesis it was designed to test, and no claim of predictive power is made on this evidence.

6. Limitations

Resolution, not stochasticity. An initial assessment attributed instability to sampling noise. Direct measurement refuted this: nine repeated runs across three transcripts returned identical scores in every case. The earlier impression arose from comparing runs conducted under different prompt revisions. Observed movements therefore reflect the transcripts rather than sampling variance. The genuine deficiency is resolution: scores occupy five values and no call in the corpus received a negative score.

Sample size. Eight holdings over three quarters at one point in the market cycle. No result here generalises.

Selection. Holdings are large-capitalisation names selected partly for transcript availability. Thinly covered companies, where such a tool would be most valuable precisely because few analysts read those calls, are where coverage fails.

Event-time approximation. The provider supplies fiscal period labels but no call dates, so entry timing is approximated. Several weeks of error is material for a ninety-day holding period.

Unadjusted returns. Returns are not benchmark-adjusted. A holding that appreciated alongside a rising index is uninformative about the earnings call.

Absence of ground truth. Themes were verified against source text, but no human-labelled corpus exists against which the absolute accuracy of a given score could be assessed.

7. Conclusions and Future Work

A functioning pipeline was delivered: portfolio definition in, verified sentiment trends and a cross-portfolio thematic view out, with every stage independently checked. The thematic component produced a usable signal. The sentiment component produced a usable interface over a measurement too coarse to support the inferential claim it was built to test.

Four directions follow, ordered by expected value:

1. **Comparative rather than absolute scoring.** Supplying the prior quarter's transcript and eliciting a judgement of relative change should address the resolution deficiency at its source, since language models are substantially more reliable at relative than absolute assessment.

2. **Expanded sample.** Thirty or more holdings across eight quarters, providing sufficient variance for the correlation analysis to constitute a genuine test.

3. **Benchmark adjustment.** Returns measured against a sector index rather than in absolute terms.

4. **Human-labelled validation set.** Twenty manually scored calls, permitting measurement of model agreement with human judgement.

Source code, verification scripts and extracted data: https://github.com/gregorykhatchikian-rgb/PortfolioVantage
Implementation: Python 3.12, pandas, Llama 3.1 70B Instruct via NVIDIA NIM, AlphaVantage, Streamlit, yfinance.